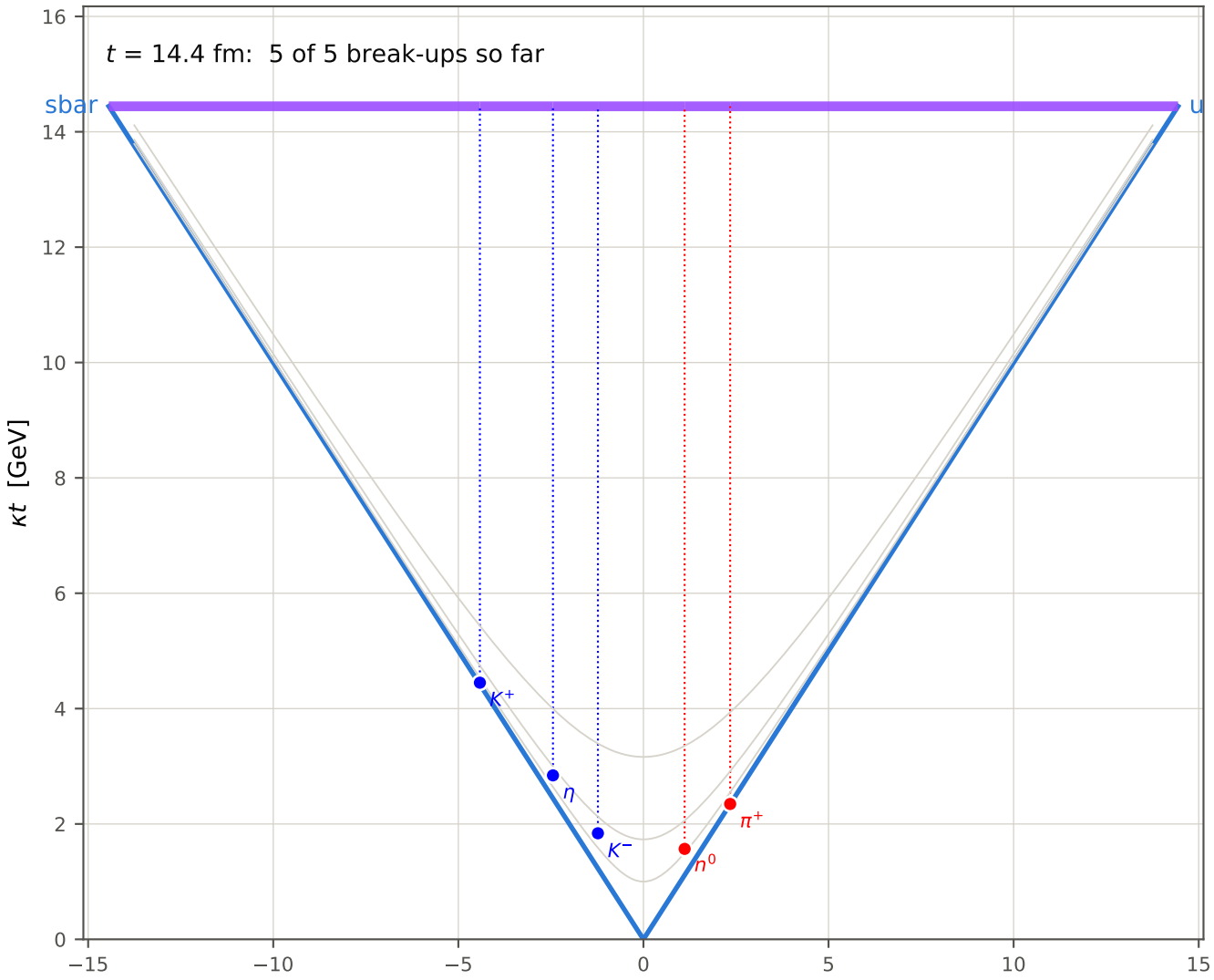
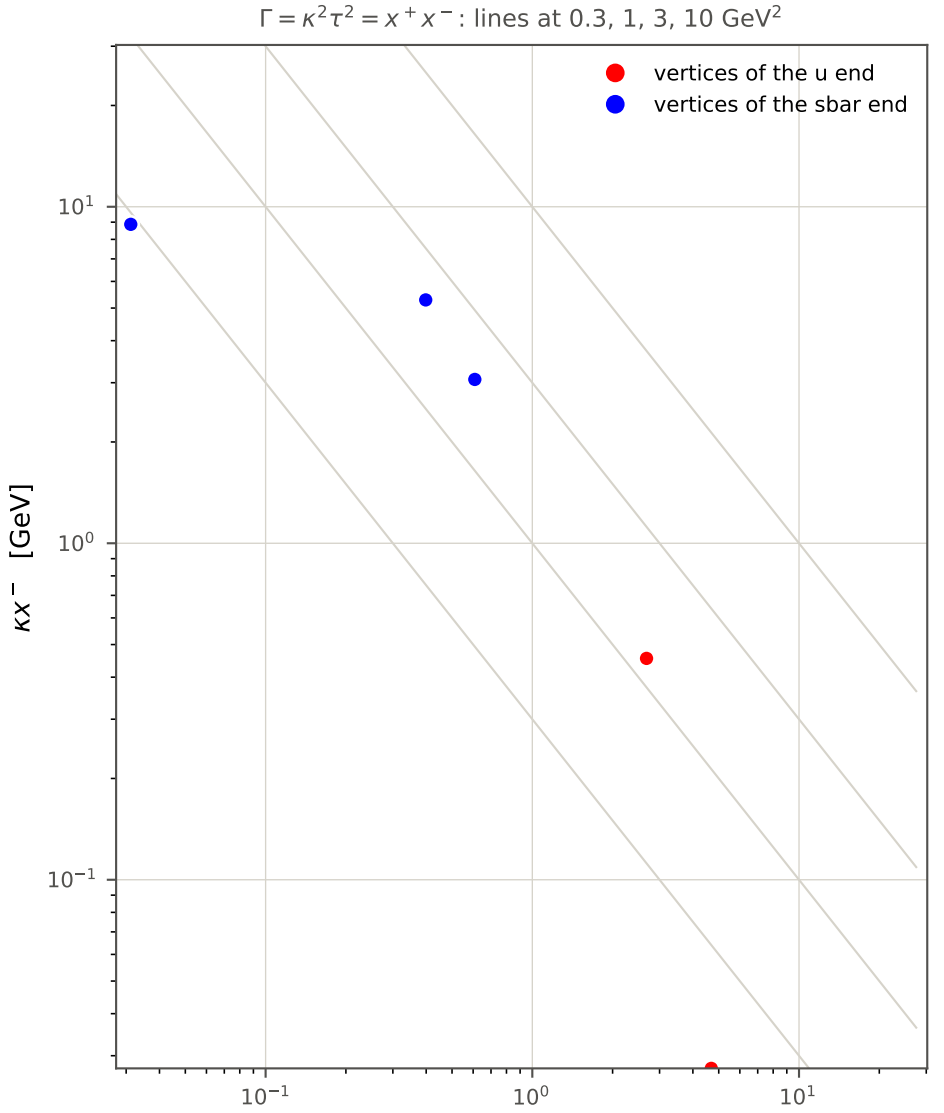
Scene 1: a string breaks: the two-parton string in its rest frame

standalone_hadronisation.py

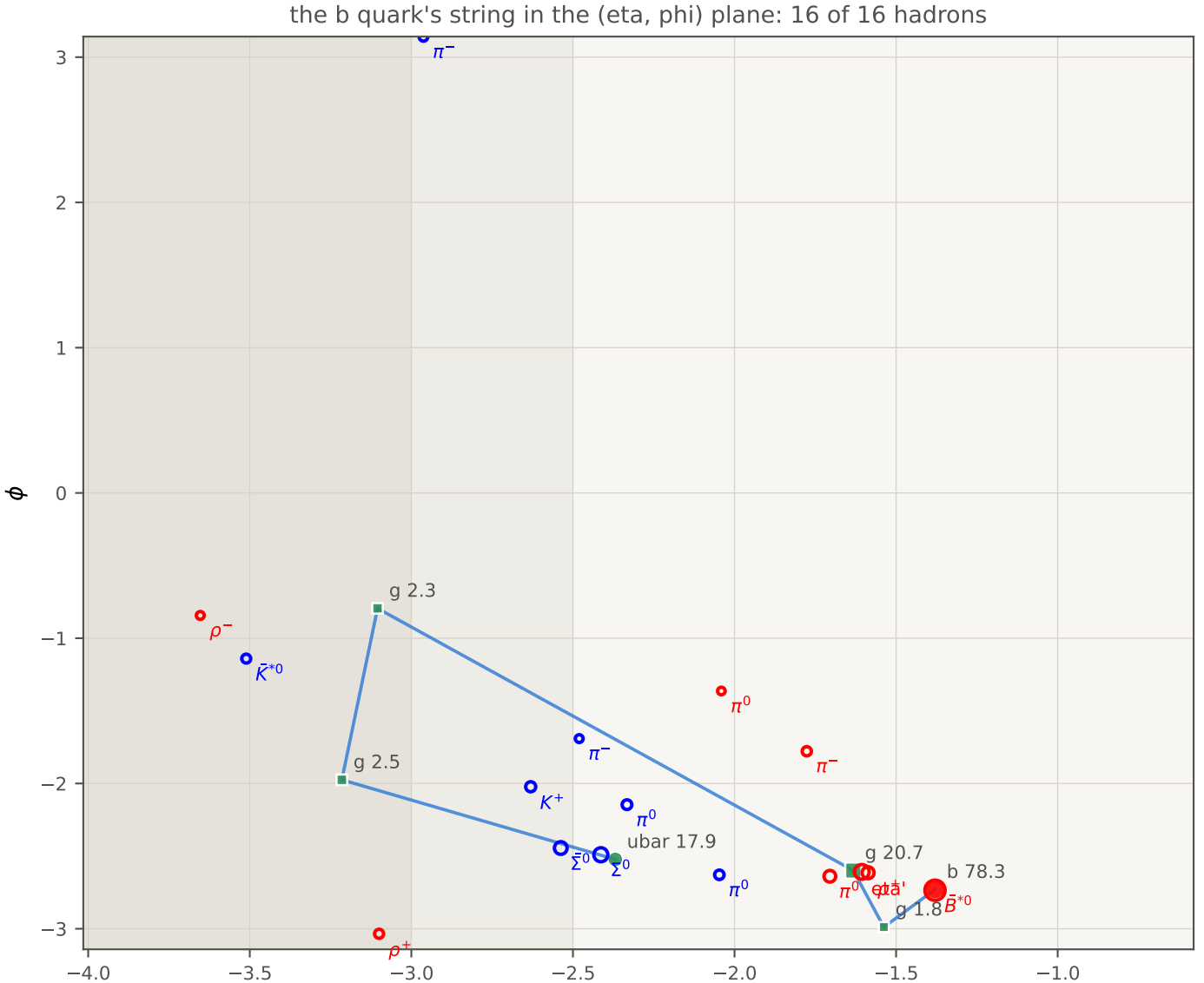




A 28 GeV u sbar string of our event in its own rest frame (seed 1). The quark and the antiquark fly apart at the speed of light, the string between them stores 1 GeV per fermi, and it breaks where the record's vertices are: every piece between two neighbouring vertices is one of the 7 hadrons. On the right the same vertices in light-cone coordinates: the area law puts them near $\Gamma = x^+x^- \sim$ 1-3 GeV^2.

Scene 2: the b quark's string: hadrons from both ends

standalone_hadronisation.py



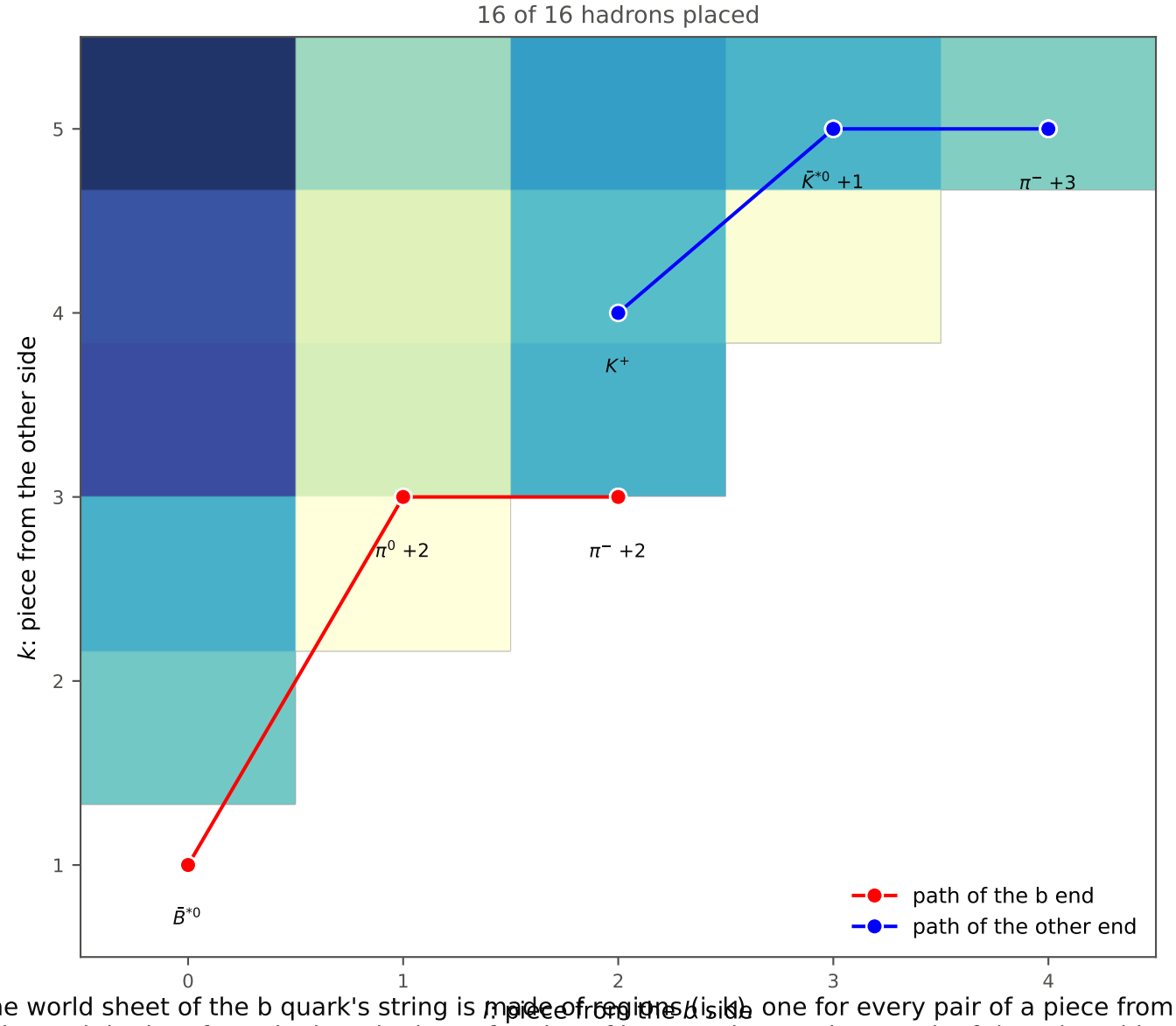
hadron	end	z	remaining W -> W_min
pi0	-	0.134	69.3 > 7.0
B*bar0	+	0.984	68.8 > 6.0
Sigmabar0	-	0.271	29.0 > 2.4
Sigma0	-	0.764	26.0 > 3.5
pi-	-	0.333	21.4 > 2.5
eta'	+	0.416	20.3 > 2.6
pi0	-	0.591	13.9 > 2.3
K*bar0	-	0.446	13.2 > 2.1
K+	-	0.565	10.4 > 2.8
rho-	+	0.314	8.8 > 2.2
pi0	+	0.242	6.8 > 2.3
rho+	+	0.444	6.5 > 2.8
pi0	+	0.603	4.5 > 2.1
pi-	+	0.274	2.4 > 2.1
rho+	+	0.549	1.2 > 3.0
pi-	-	final	1.2 > 3.0

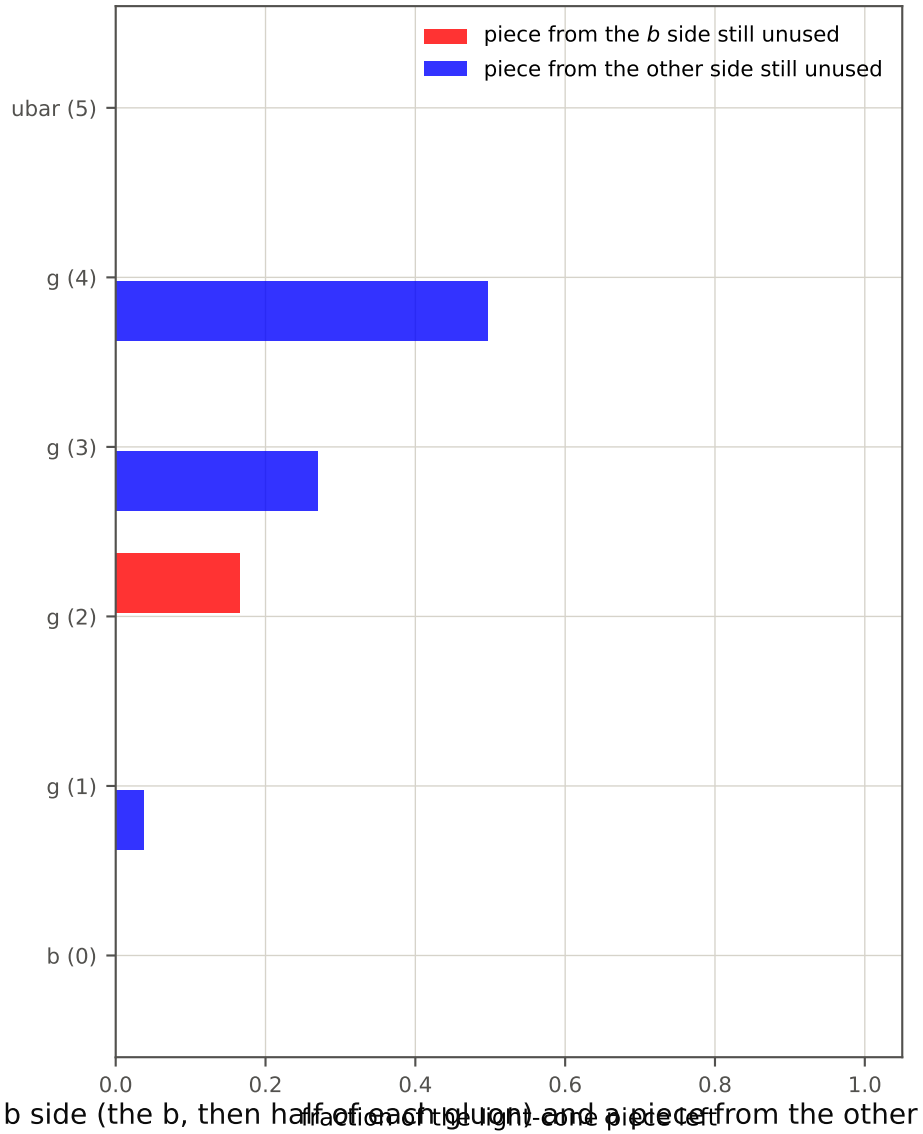
16 hadrons; momentum conserved to 1e-12 GeV;
the B*bar0 carries 100% of the b quark's energy

The string that holds the b quark: 6 partons (the b, 4 gluons and the other end), W = 69 GeV. Hadrons are produced from the two ends in random order; each takes a fraction z of the remaining light-cone momentum of its end (Lund-Bowler for the b: z = 0.984 gives the B*bar0 with x_E = 1.004) until the remaining mass drops below W_min and the last two hadrons share what is left.

Scene 3: gluons as kinks: the two ends walk through the regions of the world sheet

standalone_hadronisation.py

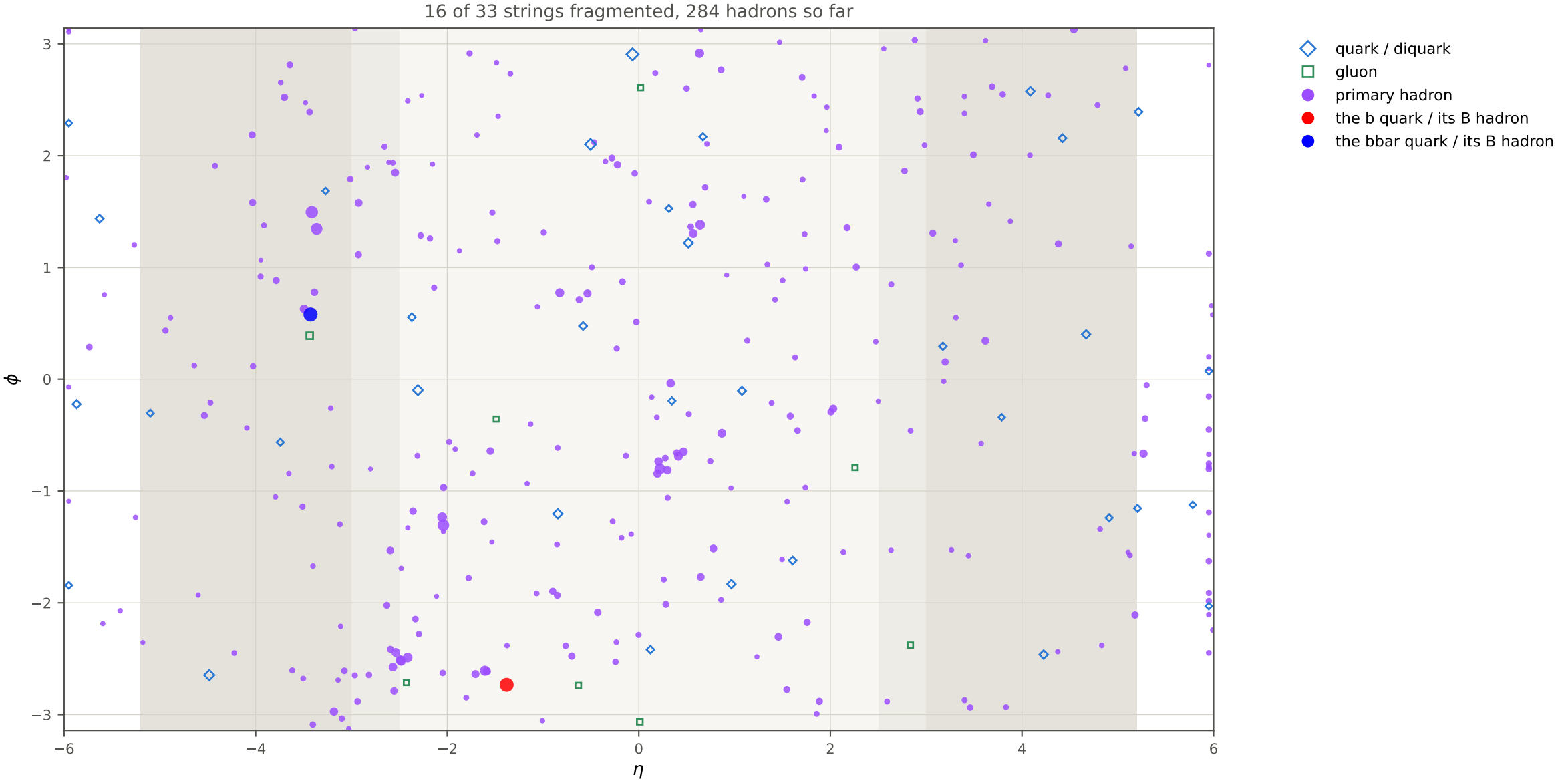




The world sheet of the b quark's string is made of regions (i, k), one for every pair of a piece from the b side (the b, then half of each gluon) and a piece from the other side. Each hadron from the b end takes a fraction of its own piece and as much of the other side as its mass needs; when a region is used up the end steps to the next one, swallowing the rest of the gluon half, which is how a hadron forms across a kink. The map darkens with the region's mass.

Scene 4: the whole event: 33 strings become 365 hadrons

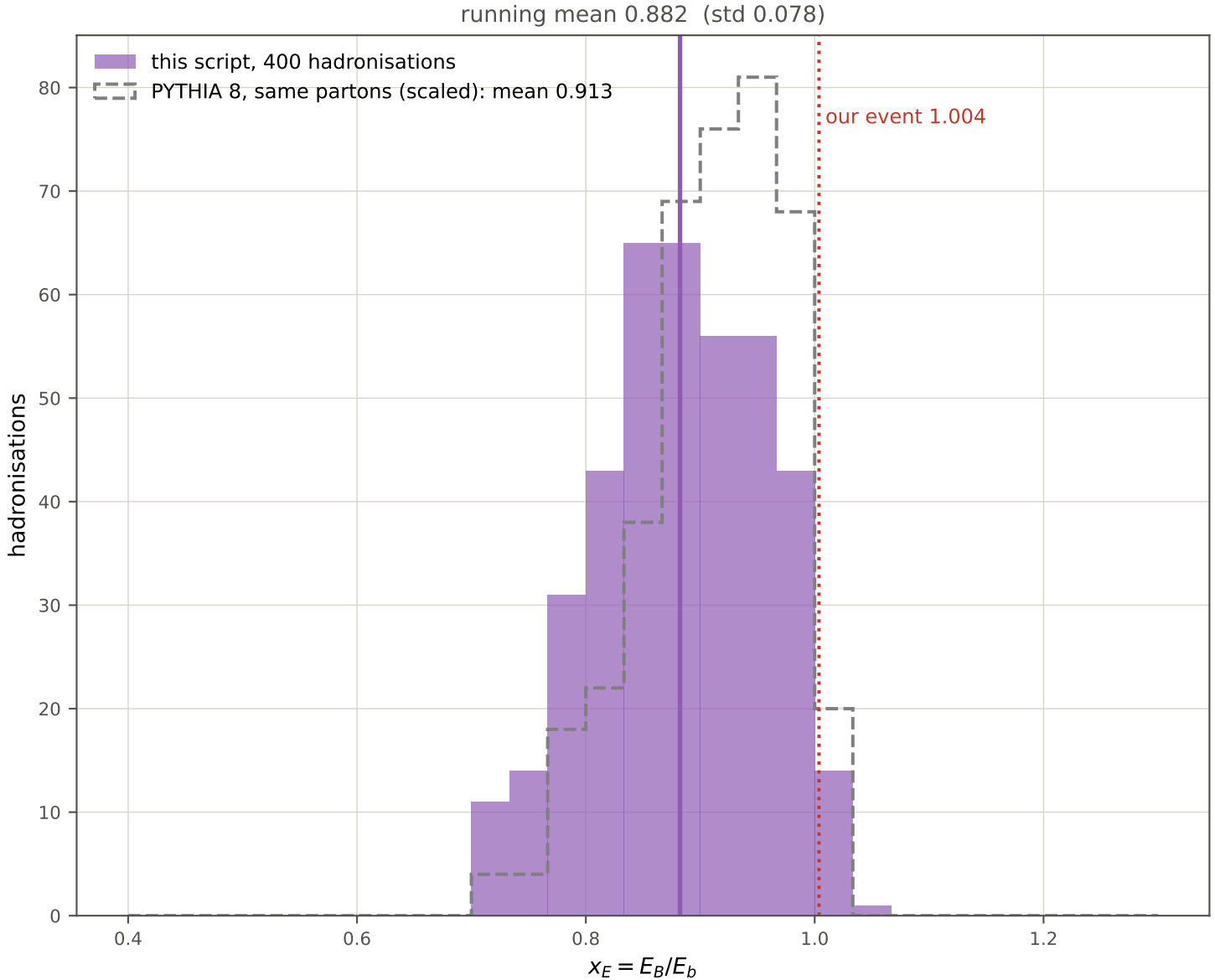
standalone_hadronisation.py

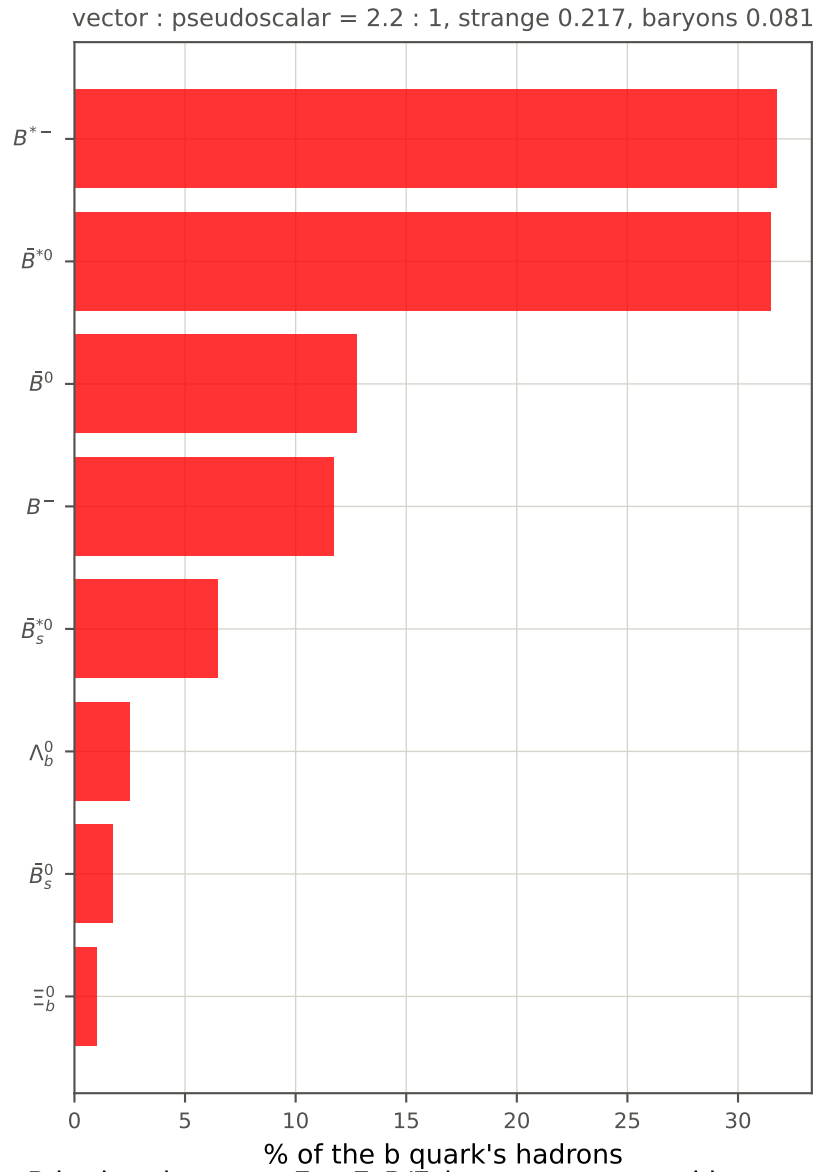


The 211 partons of stage 3 are grouped into 33 colour singlets by their colour tags (4 of them ministrings). String by string, heaviest first, the partons go and the hadrons come: 365 primary hadrons, 184 of them charged, the momentum sum unchanged. The B hadrons of the two b quarks are the red and blue dots.

Scene 5: the same partons hadronised again and again: x_E of the b quark's hadron

standalone_hadronisation.py





The partons of stage 3 are kept and only the hadronisation is redrawn, seeds 1, 2, 3, ...: the energy fraction the B hadron keeps, x_E = E_B/E_b, scatters around its Lund-Bowler mean; PYTHIA 8 on the very same partons (400 times) gives the dashed line. Our event (seed 1) is the first entry. Right: which B hadron it was.